TECHNICAL GUIDE

# Box3D Metal Architecture

A compact guide to the Apple Silicon command graph, shared-memory ownership, constraint colors, and deterministic overflow.

APPLE SILICON / METAL COMPUTE / BOX3D OVERLAY

# One ordered command graph

Supported constrained worlds encode every substep into one Metal command buffer, wait once, and read body state back once. This minimizes ownership changes on Apple unified memory.

| Phase | Stage | Work |
|---|---|---|
| 1 | Integrate velocity | Forces, damping, gyroscopic term, speed limits |
| 2 | Warm start | Ordered overflow, then conflict-free graph colors |
| 3 | Solve | Biased contact and supported-joint iterations |
| 4 | Integrate position | Position and normalized quaternion update |
| 5 | Relax | Unbiased constraint iterations |
| 6 | Restitution | Eligible convex and mesh contacts |

Unconstrained worlds use a smaller fused path. Unsupported constrained worlds keep the CPU solver and may use only the independent Metal position stage.

Unified memory is not free synchronization

Persistent shared buffers avoid PCIe copies, but command submission, cache coherence, residency, and CPU/GPU ownership transitions still determine crossover.

# Data layout and ownership

- Body state and compact body properties use geometrically grown persistent buffers.
- CPU contact preparation writes directly into shared Metal constraint allocations.
- Distance and parallel joints use separate compact type-dense buffers.
- Only accumulated joint solver state is unpacked after the command buffer.
- C/Metal size and offset assertions guard every shared ABI.
- Scalar shader vectors avoid native float3 padding surprises.

## Graph colors

Constraints inside one non-overflow color never write the same dynamic body, so one GPU thread can own one constraint without atomics. Colors remain ordered with buffer barriers.

## Deterministic overflow

Overflow constraints may share bodies. A single Metal thread walks them in upstream order. Mixed distance/parallel overflow uses an eight-byte type/index descriptor, preserving order with one launch per phase.

# Apple GPU implementation choices

| Choice | Reason |
|---|---|
| One command buffer per step | Amortize submission and synchronization |
| Shared persistent storage | Exploit unified memory without redundant contact copies |
| Pipeline-derived group width | Match execution width and occupancy limits |
| No atomics in colored path | Conflict graph provides exclusive body writes |
| Serial overflow kernel | Correctness for body-sharing constraints |
| Safe Metal math | Reduce avoidable drift from the CPU oracle |

Telemetry. The profile reports device, threshold, dispatch/fallback counters, and latest GPU time for position, unconstrained, contact, and joint paths. Treat these as routing evidence, not as a whole-engine GPU percentage.